EE 121B: Principles of Semiconductor Device Design

Lecture 9:

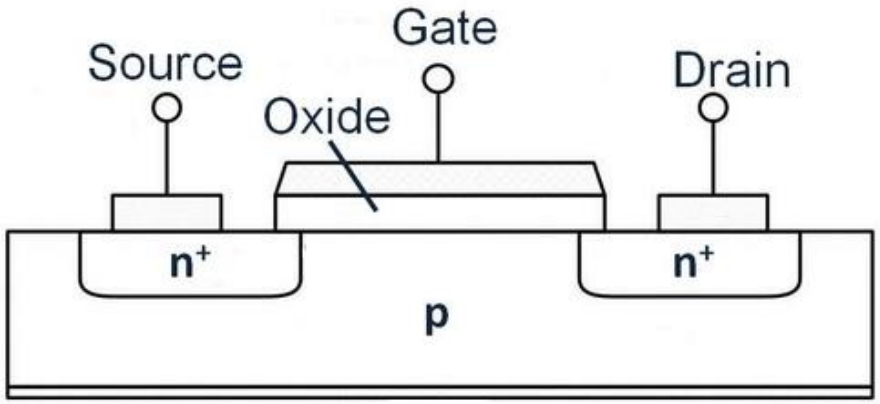
pn Junction Diodes

Sam Emaminejad

Course Overview

1. Semiconductor Properties
2. Metal-Semiconductor Contacts
3. P-N Junction
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



http://qph.is.quoracdn.net/main-qimg-38a99b4f215e161bf31964795cfce7d2?convert_to_webp=true

Lecture 9

OUTLINE

- pn Junction Diodes (cont'd)
  - Charge control model
  - Small-signal model
  - Transient response: turn-off
  - Transient response: turn-on

Reading: Pierret 7, 8.1; Hu 4.10-4.11

Minority-Carrier Charge Storage

- Under forward bias ($V_A > 0$), excess minority carriers are stored in the quasi-neutral regions of a pn junction:

p-side

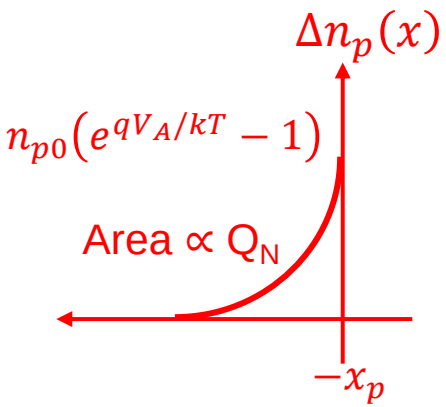
$$Q_N = -qA\int_{-x_p}^{-\infty} \Delta n_p(x)dx$$

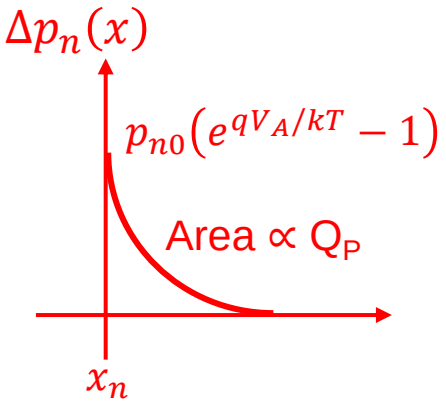
$$= -qA\Delta n_p(-x_p)L_N$$

n-side

$$Q_P = qA\int_{x_n}^{\infty} \Delta p_n(x)dx$$

$$= qA\Delta p_n(x_n)L_P$$





Derivation of Charge Control Model

Consider the **n quasi-neutral region** of a forward-biased pn junction:

•The minority carrier diffusion equation is (assuming G_L=0):

$$\frac{\partial \Delta p_n}{\partial t} = D_P \frac{\partial^2 \Delta p_n}{\partial x^2} - \frac{\Delta p_n}{\tau_p}$$

•Since the electric field is very small, $J_P = -qD_P \frac{\partial \Delta p_n}{\partial x}$

•Therefore $\frac{\partial (q\Delta p_n)}{\partial t} = -\frac{\partial J_P}{\partial x} - \frac{q\Delta p_n}{\tau_p}$

The number of holes decreases due to recombination with (majority carrier) electrons, at a rate of proportional to the excess hole concentration

Rate of change in excess hole charge

If the hole current entering a volume Adx is greater than the hole current leaving the volume, then the number of holes in the volume increases over time

Derivation Assuming a Long Base

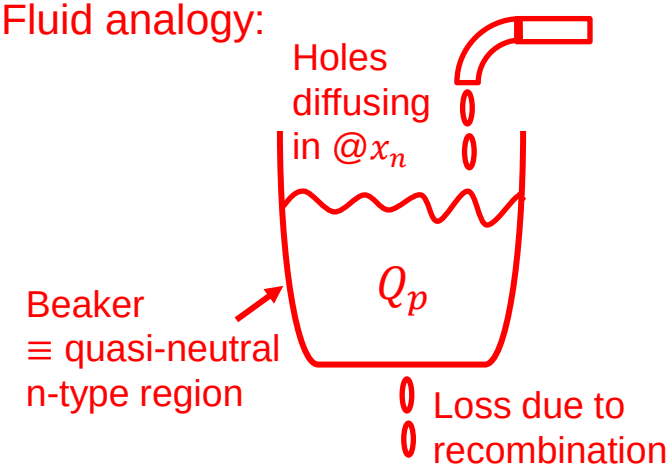
- Integrating over the n quasi-neutral region:

$$\frac{\partial}{\partial t}\left[qA\int_{x_n}^{\infty} \Delta p_n dx \right] = -A\int_{J_p(x_n)}^{J_P(\infty)} dJ_P - \frac{1}{\tau_p}\left[qA\int_{x_n}^{\infty} \Delta p_n dx \right]$$

- Note that $-A\int_{J_p(x_n)}^{J_P(\infty)} dJ_P = -AJ_P(\infty) + AJ_P(x_n) = AJ_P(x_n) = I_P(x_n)$ ($J_P(\infty) = 0$)

- So $\frac{dQ_P}{dt} = I_P(x_n) - \frac{Q_P}{\tau_p}$ = 0 in steady state

Fluid analogy:



Charge Control Model

We can calculate pn-junction current in 2 ways:

1. From slopes of $\Delta n_p(-x_p)$ and $\Delta p_n(x_n)$
2. From steady-state charges Q_N, Q_P stored in each excess-minority-charge distribution:

$$\frac{dQ_P}{dt} = I_P(x_n) - \frac{Q_P}{\tau_p} = 0$$

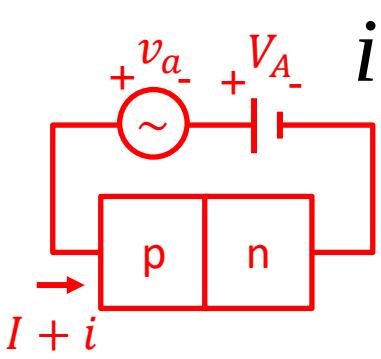
$$\Rightarrow I_P(x_n) = \frac{Q_P}{\tau_p}$$

$$\text{Similarly, } I_N(-x_p) = \frac{-Q_N}{\tau_n}$$

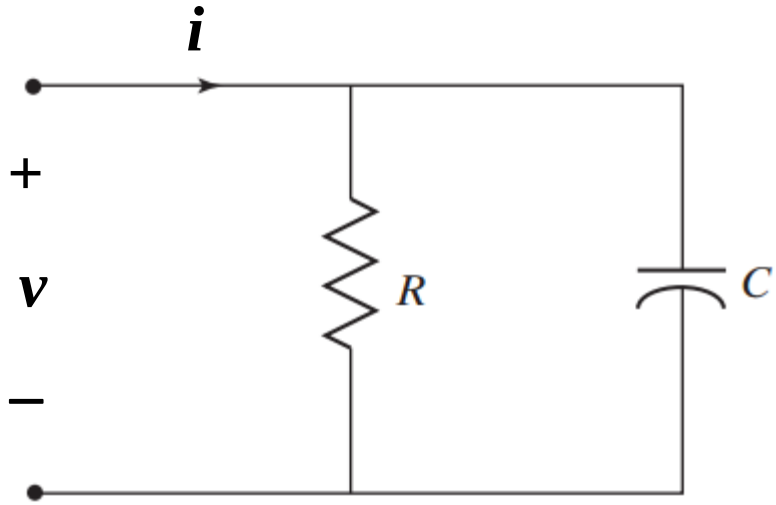
$$I = I_P(x_n) + I_N(-x_p) = \frac{Q_p}{\tau_p} - \frac{Q_n}{\tau_n}$$

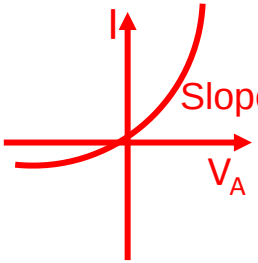
Small-Signal Model of the Diode

Under forward bias:



$$i = \frac{v_a}{R} + C\frac{dv_a}{dt}$$





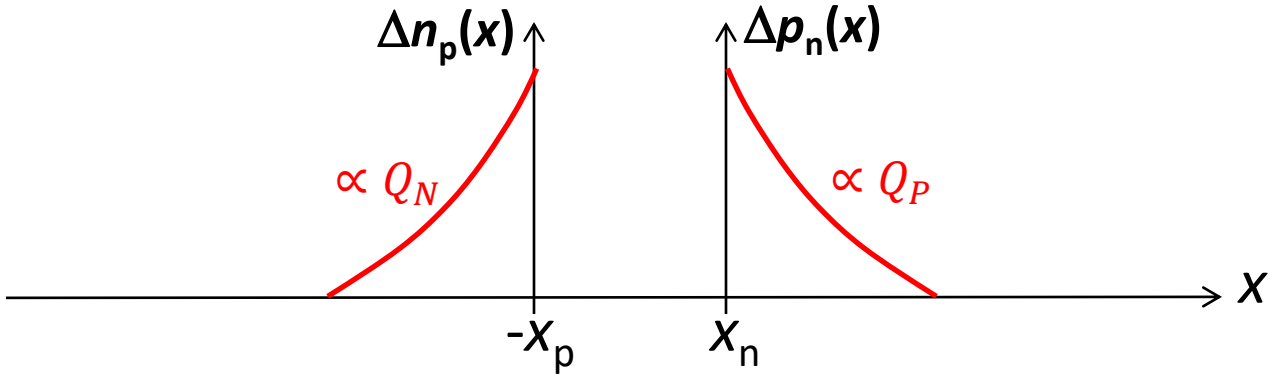
$$\text{Slope} = \frac{1}{R} = \frac{dI_{DC}}{dV_A} = \frac{d}{dV_A} I_0\left(e^{qV_A/kT} - 1\right) \approx \frac{d}{dV_A} I_0 e^{qV_A/kT} = \frac{q}{kT}\underbrace{I_0 e^{\frac{qV_A}{kT}}}_{\sim I_{DC}}$$

Small-signal conductance:

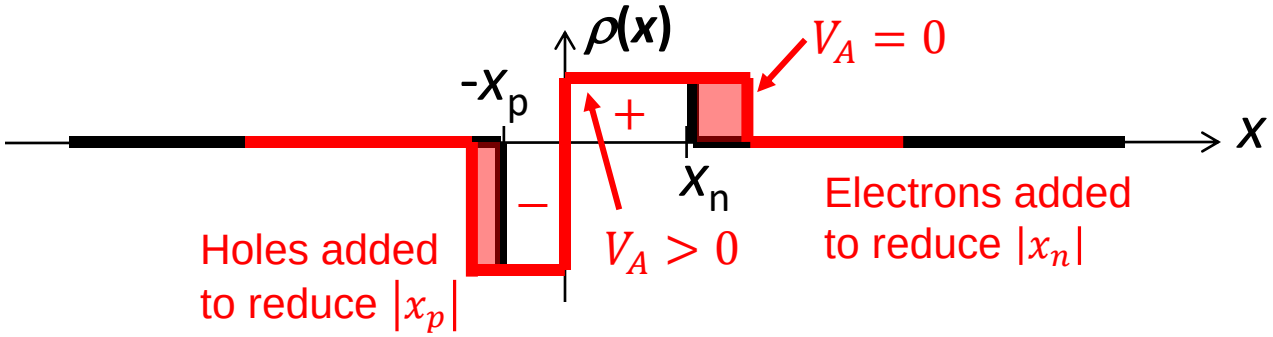
$$G \equiv \frac{1}{R} = \frac{q}{kT} I_0 e^{qV_A/kT} \cong \frac{I_{DC}}{kT/q}$$

Charge Storage in pn Junction Diode

- Excess minority carriers in the quasi-neutral region:



- Majority carriers stored at the edges of the depletion regions:

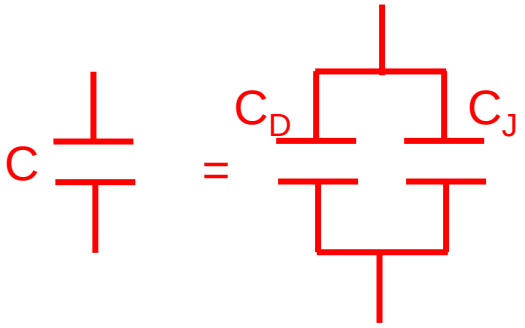


pn Junction Small-Signal Capacitance

2 types of capacitance associated with a pn junction:

depletion capacitance $C_J \equiv \left| \frac{dQ_{dep}}{dV_A} \right|$

– due to variation of depletion charge



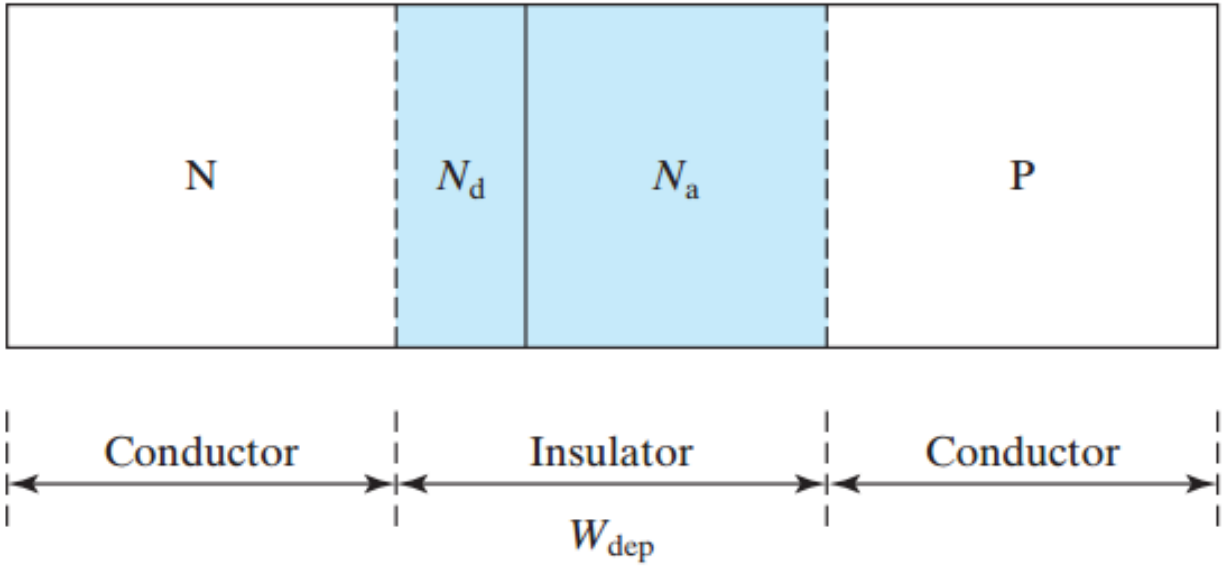
diffusion capacitance $C_D \equiv \left| \frac{dQ}{dV_A} \right|$

–due to variation of stored minority charge in the quasi-neutral regions

For a one-sided p^+n junction $Q = Q_P + Q_N \cong Q_P$ so

$$C_D = \left| \frac{dQ_P}{dV_A} \right| = \tau_p \frac{dI}{dV_A} = \tau_p G = \frac{\tau_p I_{DC}}{kT/q}$$

Depletion Capacitance



C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 4-8

$$C_J \equiv \left| \frac{dQ_{dep}}{dV_A} \right| = A \frac{\varepsilon_s}{W}$$

What are three ways to reduce C_J?

1) Lower the dopant concentration (N_A, N_D)
2) Reduce junction area
3) Apply reverse bias

Total pn-Junction Capacitance

$$C = C_D + C_J$$

$$C_D = \frac{\tau I_{DC}}{kT/q} \propto \left(e^{qV_A/kT} - 1\right)$$

Increases exponentially with forward bias

$$C_J = A\frac{\varepsilon_s}{W}$$

← Increases gradually with forward bias

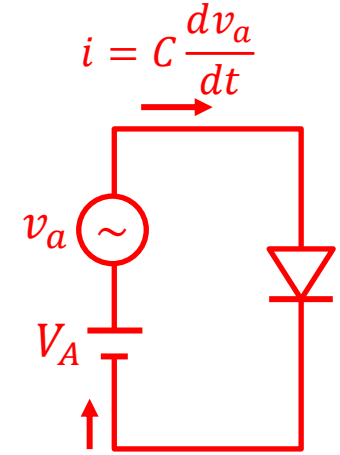
- C_D dominates at moderate to high forward biases
- C_J dominates at low forward biases, reverse biases

Using *C-V* Data to Determine Doping

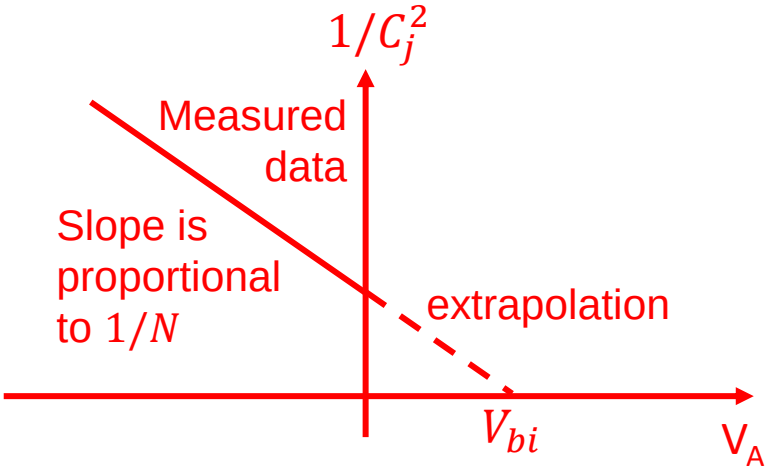
Under reverse bias, for a one-sided junction:

$$\frac{1}{C_J^{\,2}} = \frac{W^2}{A^2\varepsilon_s^{\,2}} \cong \frac{2(V_{bi} - V_A)}{A^2 q \varepsilon_S N}$$

Dopant concentration on the more lightly doped side



Reverse bias
→ DC current is very small



1) N can be determined from slope
2) V_{bi} can be determined from x-intercept
3) From (1) and (2), dopant concentration on heavily doped side can be determined

Example

If the slope of the $(1/C)^2$ *vs.* V_A characteristic is $-2\times10^{23}\ F^{-2}\ V^{-1}$, the intercept is 0.84V, and A is 1 μm^2, find the dopant concentration N_l on the more lightly doped side and the dopant concentration N_h on the more heavily doped side.

Solution:

$$N_l = 2/(slope \times q\varepsilon_s A^2)$$

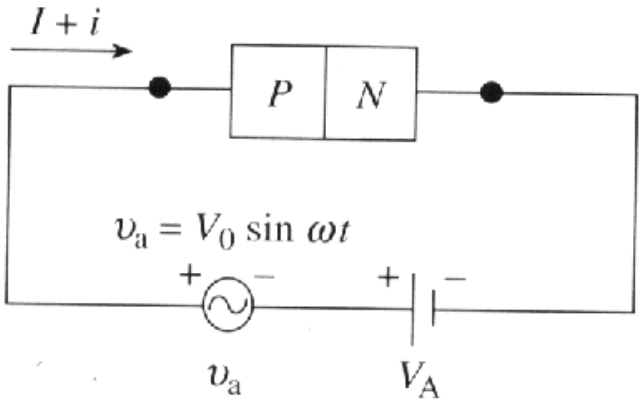
$$= 2/(2\times10^{23}\times1.6\times10^{-19}\times10^{-12}\times\left(10^{-8}\right)^2)$$

$$= 6\times10^{15}\ \text{cm}^{-3}$$

$$V_{bi} = \frac{kT}{q}\ln\frac{N_h N_l}{n_i^{\,2}} \quad\Rightarrow\quad N_h = \frac{n_i^{\,2}}{N_l}e^{\frac{qV_{bi}}{kT}} = \frac{10^{20}}{6\times10^{15}}e^{\frac{0.84}{0.026}} = 1.8\times10^{18}\ \text{cm}^{-3}$$

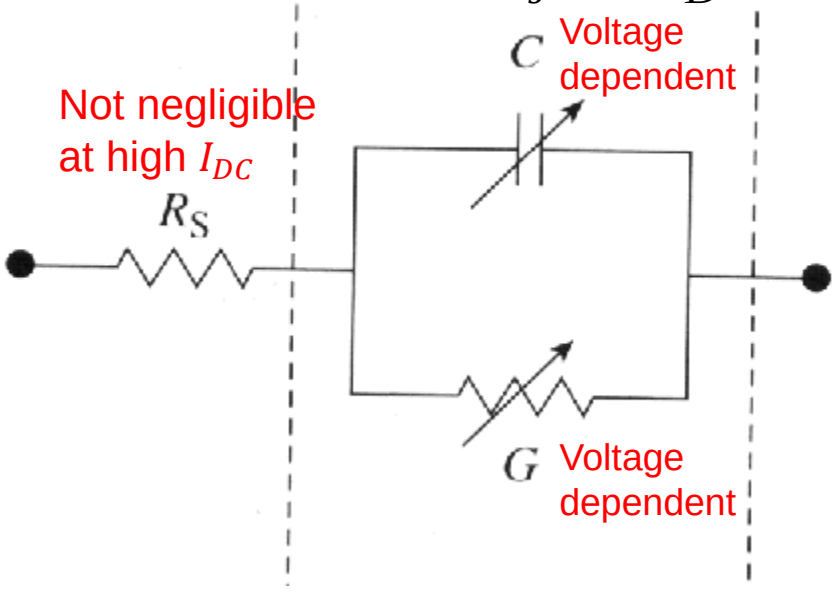
Assumes both sides are non-degenerately doped, so that the Boltzmann approximation is valid

Small-Signal Model Summary



$$I_{DC} = I_0(e^{qV_A/kT} - 1)$$

$$C = C_J + C_D$$



Depletion capacitance $C_J = \dfrac{A\varepsilon_s}{W}$

Conductance $G \cong \dfrac{I_{DC}}{kT/q}$

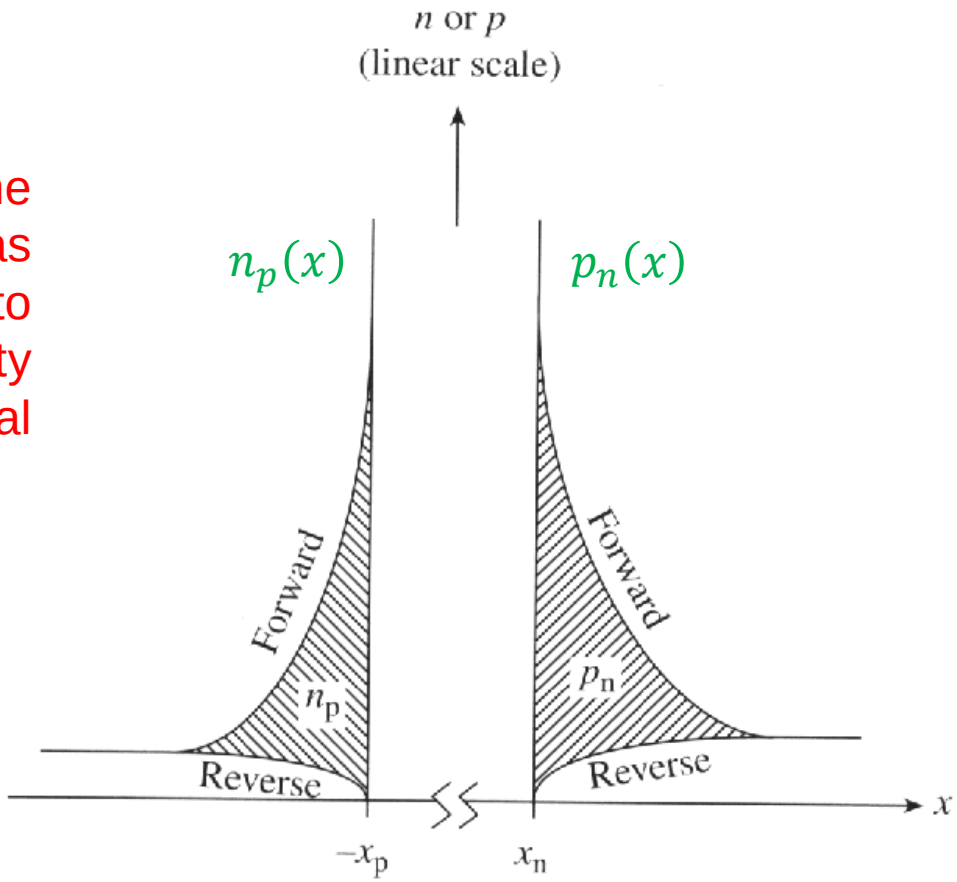
Diffusion capacitance $C_D = \dfrac{\tau I_{DC}}{kT/q}$

R. F. Pierret, *Semiconductor Device Fundamentals*, p. 302

Transient Response of pn Diode

- Suppose a pn-diode is forward biased, then suddenly turned off at time $t = 0$. Because of C_D, the voltage across the pn junction depletion region cannot be changed instantaneously.

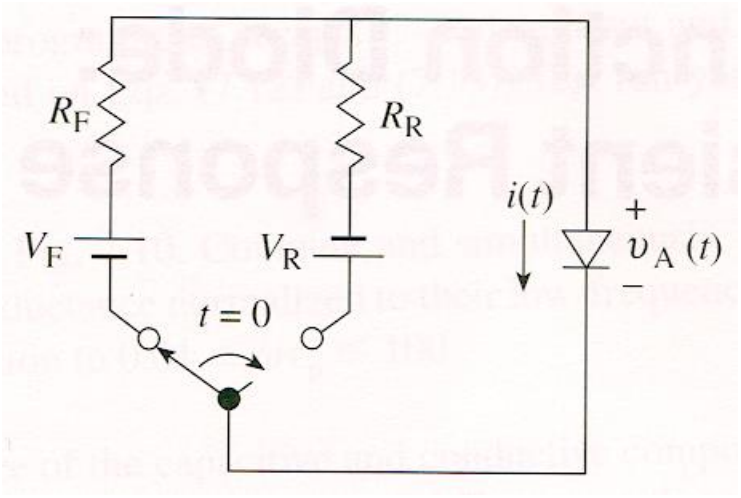
The time delay in switching between the FORWARD-bias and REVERSE-bias states is due to the time required to change the amount of excess minority carriers stored in the quasi-neutral regions.

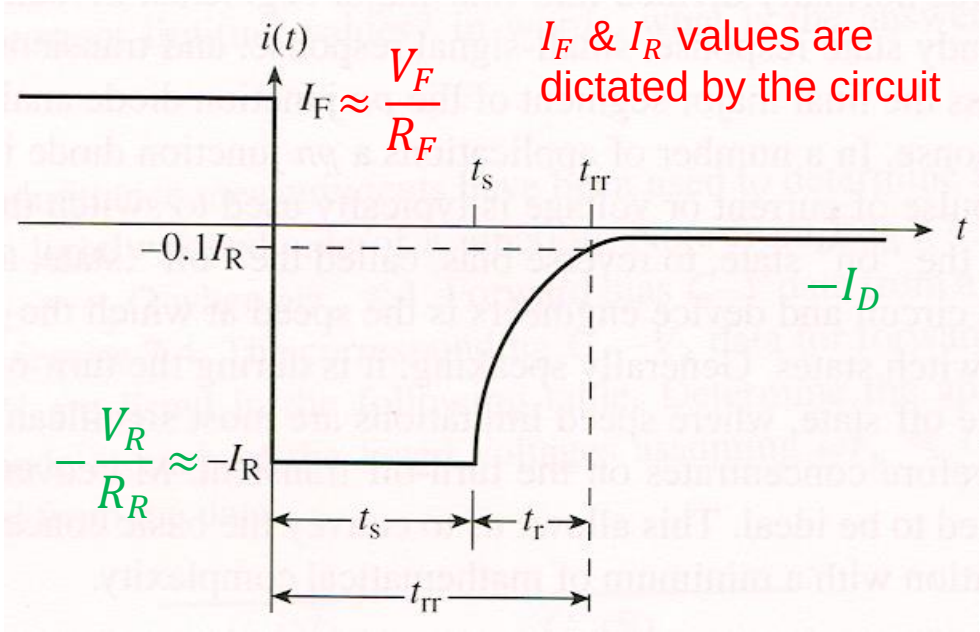


R. F. Pierret, *Semiconductor Device Fundamentals*, Fig. 8.2

Turn-Off Transient

- In order to turn the diode off, the excess minority carriers must be removed by net carrier flow out of the quasi-neutral regions and/or recombination
  - Carrier flow is limited by the switching circuitry

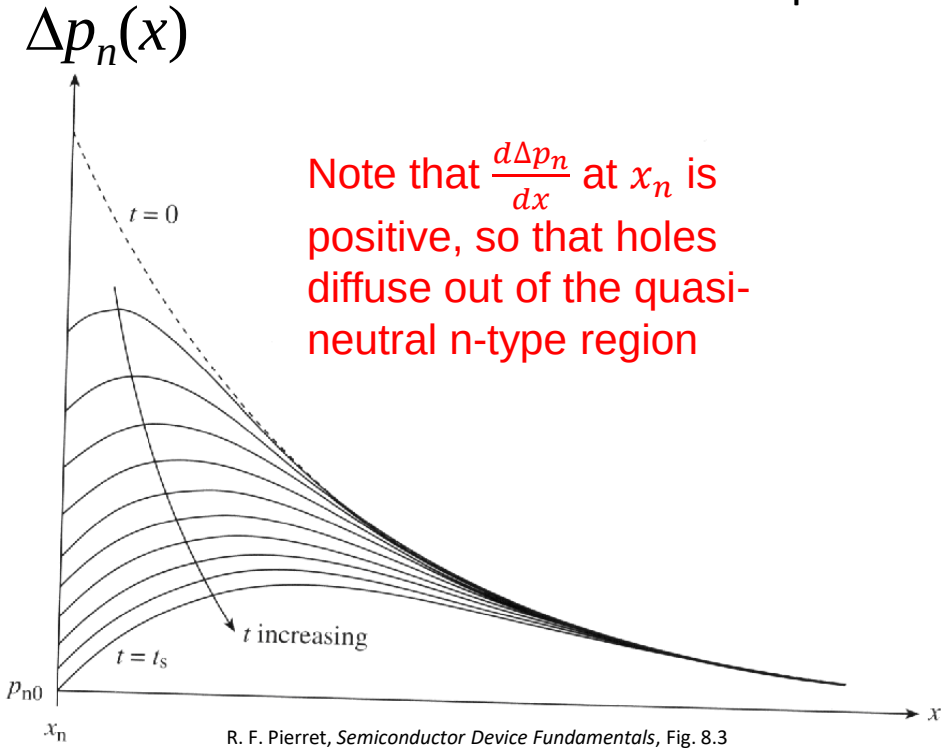




R. F. Pierret, *Semiconductor Device Fundamentals*, p. 328

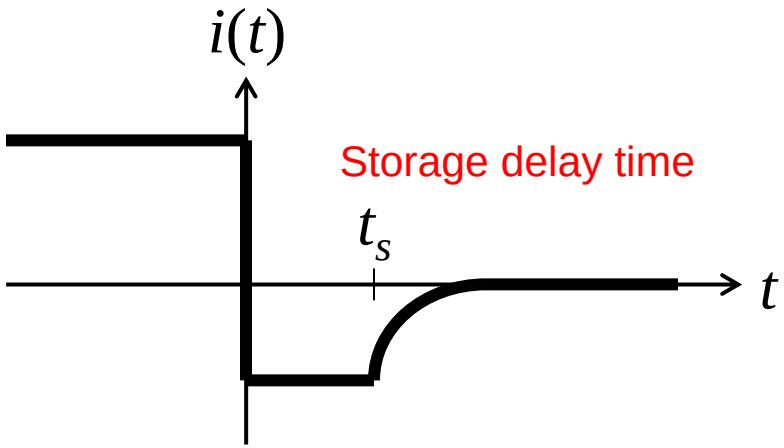
Decay of Stored Charge

Consider a **p⁺n diode** ($Q_p >> Q_n$):

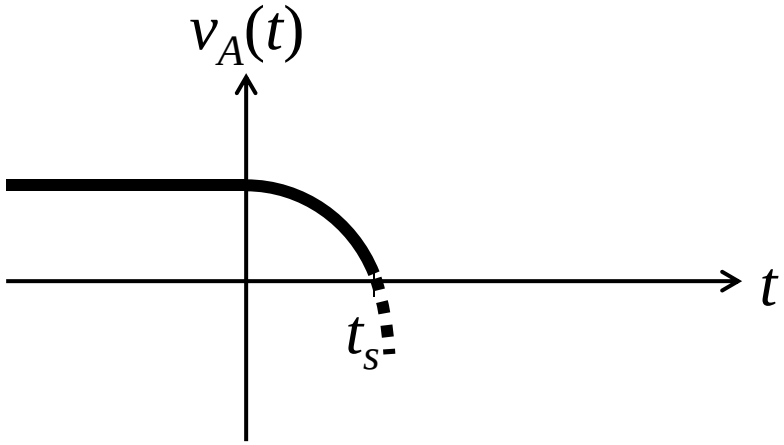


R. F. Pierret, *Semiconductor Device Fundamentals*, Fig. 8.3

Note that $\frac{d\Delta p_n}{dx}$ at x_n is positive, so that holes diffuse out of the quasi-neutral n-type region



Storage delay time



For $t > 0$:

$$\left.\frac{dp_n}{dx}\right|_{x=x_n} = -\frac{i}{qAD_p} > 0$$

Storage Delay Time, t_s

- t_s is the primary "figure of merit" used to characterize the transient response of pn junction diodes

$$\frac{dQ_p}{dt} = i - \frac{Q_p}{\tau_p} = -\left(I_R + \frac{Q_p}{\tau_p} \right) \qquad 0^+ \le t \le t_s$$

- By separation of variables and integration from $t = 0^+$ to $t = t_s$, noting that $I_F = Q_p(t=0)/\tau_p$

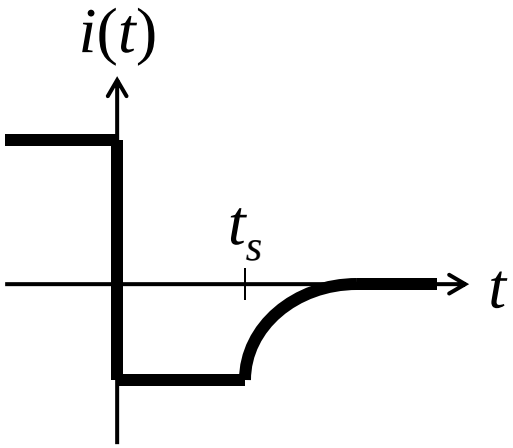
and making the approximation $Q_p(t = t_s) = 0$

We conclude that $t_s \cong \tau_p \ln\left(1 + \frac{I_F}{I_R}\right)$

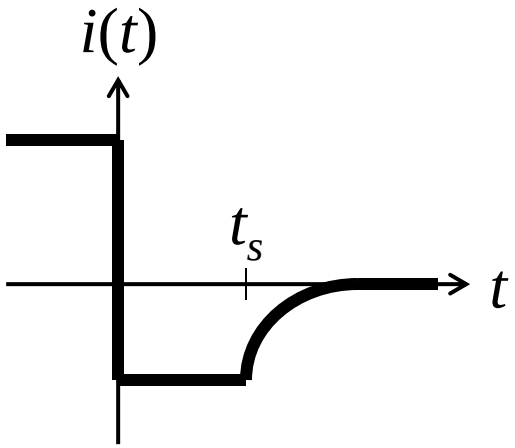
Qualitative Examples

Illustrate how the turn-off transient response would change:

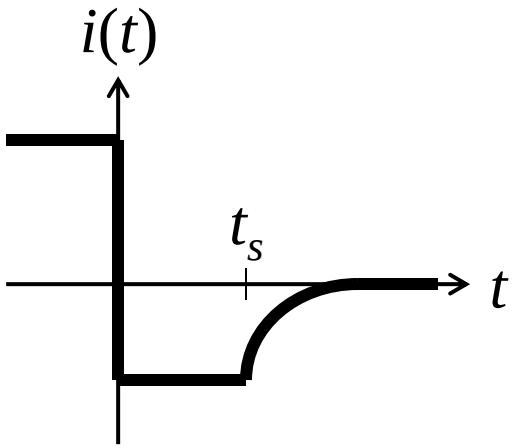
Increase I_F



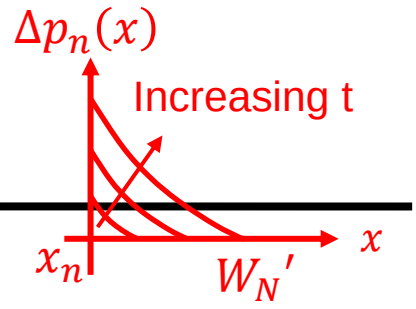
Increase I_R



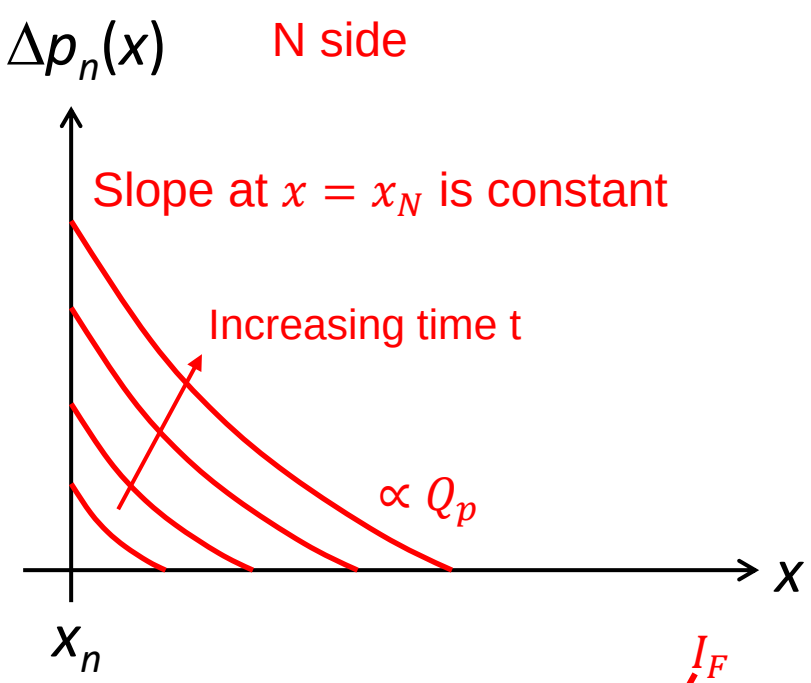
Decrease τ_p

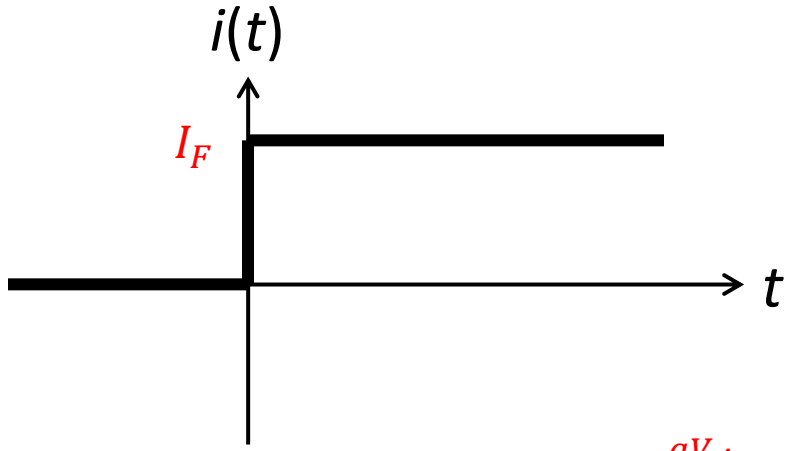


Turn-On Transient



Consider a **p⁺n diode** ($Q_p >> Q_n$):

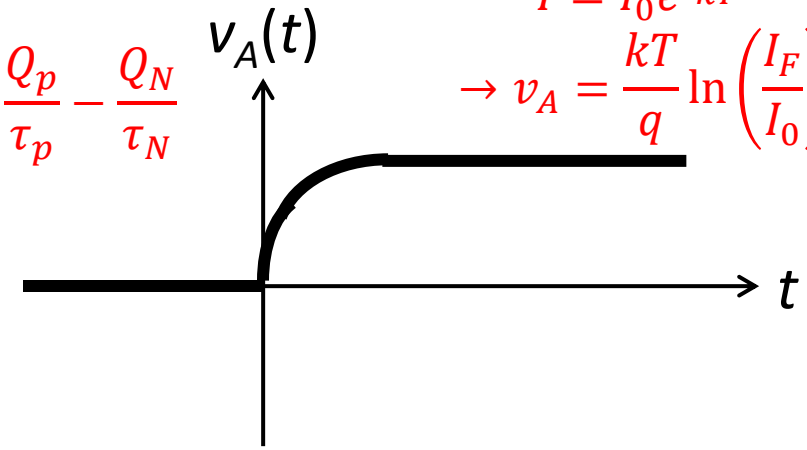




$$I = \frac{Q_p}{\tau_p} - \frac{Q_N}{\tau_N}$$

$$I = I_0 e^{\frac{qV_A}{kT}}$$

$$\rightarrow v_A = \frac{kT}{q} \ln\left(\frac{I_F}{I_0}\right)$$



For $t > 0$: I_F

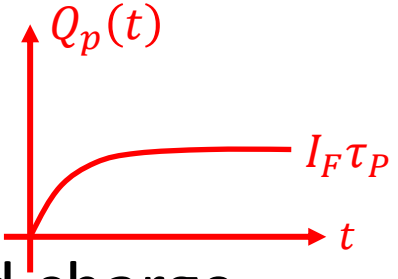
$$\left.\frac{dp_n}{dx}\right|_{x=x_n} = -\frac{i}{qAD_p} < 0$$

Charge control model for p+n junction ($Q_P \gg Q_N$)

$$\frac{dQ_p}{dt} = i - \frac{Q_p}{\tau_p} = I_F - \frac{Q_p}{\tau_p} \qquad \text{for } t \geq 0^+$$

- By separation of variables and integration, we have

$$Q_p(t) = I_F \tau_p \left(1 - e^{-t/\tau_p}\right)$$



- If we assume that the build-up of stored charge occurs quasi-statically so that

$$I = \frac{Q_p}{\tau_p} = I_0\left(e^{\frac{qV_A}{kT}} - 1\right)$$

V_A is a function of time

$$Q_p(t) = I_{diffusion}\tau_p = I_0\left(e^{qv_A/kT} - 1\right)\tau_p$$

then $$v_A(t) = \frac{kT}{q}\ln\left[1 + \frac{I_F}{I_0}\left(1 - e^{-t/\tau_p}\right)\right]$$

i.e. recombination is negligible

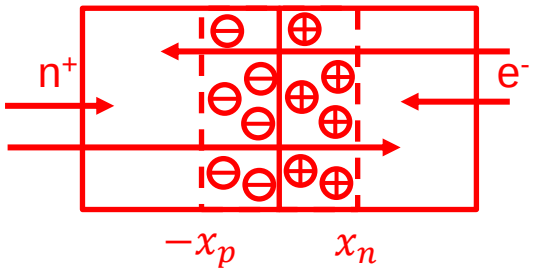
- If τ_p is large, then the time required to turn on the diode is approximately $\Delta Q/I_F$

(I_F is the rate at which charge enters the diode)

$$\text{where} \quad \Delta Q = \Delta Q_p + \Delta Q_j$$

Excess minority carrier charge stored in quasi-neutral region

Charge stored at the edges of the depletion region



Acknowledgement

- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
- Professor Tsu-Jae King Liu EE130/230A notes